

How Microcontrollers are Getting Better and Better



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A microcontroller (MCU) designed for development by software engineers is not a bad idea. Considering that the current system design team mix at many design houses has a good share of computer science engineers, this could easily bring down the barrier to entry for getting started with embedded systems design and spur creative designs.

"We are trying to shift the discussion from bits and bytes at the silicon level to the platform level, and help our customers be more efficient with their software," notes Matt Johnson, VP and GM of automotive MCUs, Freescale Semiconductor, in their press release about the new S32K MCU family. The press release claims it to be the first automotive MCU product line specifically designed to speed up and simplify software development.

How else can you speed up development? Atmel provides ATSAMD11-XPRO development kit along with their new Atmel SMART SAM DA1 MCU in order to accelerate design and development process.

MXChip's Microcontroller based Internet Connectivity Operating System (MiCO) is an Internet of Things (IoT) operating system (OS) that runs on a 32-bit MCU with middleware for IoT equipment and smart

hardware applications, with secure, low power and upward compatible features. Further, with MiCO OS developers can simplify the development process in their smart hardware development, lowering overall software development and maintenance costs, as well as bringing products faster to market.

Going beyond the traditional AUTomotive Open System Architecture (AUTOSAR) MCU Abstraction Layer (MCAL), Freescale is also introducing an automotive-grade software development kit providing critical middleware for a set of drivers necessary to operate S32K MCUs. Their S32 Design Studio provides a platform capable of supporting time-saving software and tools. These MCUs also have peripheral and package compatibility with each other, letting designers make the most of complete software reuse and reduce design cost.

New MCUs like Microchip's PIC16F18877 also come with core independent peripherals that can handle their tasks with no code or supervision from the central processing unit (CPU) to maintain operation. As a result, these simplify the implementation of complex control systems and give designers the flexibility to innovate.

Making low-power designs look easy

"The IoT and wearables are driving low-power chips with every vendor looking at different ultra-low-power benchmarks. Vendors are also trying to deliver chips that can work on harvested energy or can last on a single coin-cell battery for up to three years. This also fits into the trend of disposable devices like medical patches," explains Praveen Ganapathy, director applications-processors, sales and applications, Texas Instruments India.

While batteries for sensor based products are usually designed for a low current supply, these face a higher processing load during operations like noise elimination

The die for an Intel 8742, an 8-bit MCU

